



ग्रेटर नोएडा इंस्टीट्यूट ऑफ टेक्नोलॉजी (इंजीनियरिंग इंस्टीट्यूट)
GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)



Presentation
On
EMAIL AUTOMATION USING LLMS

Group Number: 26

Presented by

VIVEK KUMAR

VATSAL CHAUDHARY

VISHVESHVER

SHREYANSH SINGH RATHORE

Under the Supervision

DR. ASIF KHAN

(Department of CSE _GNIOT Greater Noida, India)

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1.Introduction

- Email remains one of the most widely used communication channels globally, with over 300 billion emails sent daily across professional and personal domains.
- Manual email drafting is time-consuming, repetitive, and often lacks personalization, leading to reduced productivity in enterprise environments.
- Recent advances in Large Language Models (LLMs) offer a transformative approach to intelligent email automation, enabling context-aware generation, summarization, and personalized communication at scale.
- This project proposes an AI-powered email automation platform that leverages LLM capabilities to generate professional email drafts, summarize threads, rewrite content in different tones, classify emails, and detect priority levels.

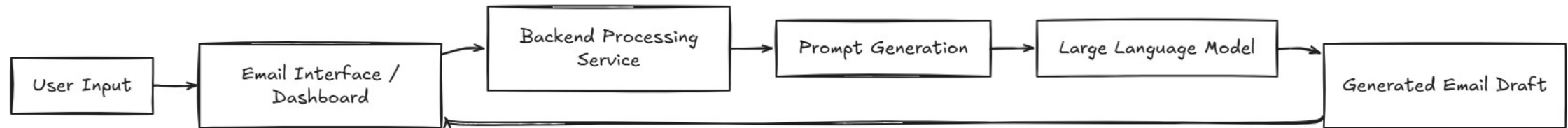
2. Research Objectives

- Reduce the time required for drafting professional emails through automated LLM-based content generation.
- Improve personalization in automated email communication by incorporating user preference memory and contextual awareness.
- Enable context-aware email generation that can summarize threads, rewrite content in different tones, classify messages, and detect priority.
- Evaluate productivity improvements achieved through LLM-based email automation compared to traditional manual drafting approaches.
- Design a modular and scalable system architecture comprising a backend processing layer, prompt construction module, LLM integration layer, and analytics and monitoring module.

3.Literature Survey

- **Transformer-Based Language Models:** The transformer architecture (Vaswani et al., 2017) introduced self-attention mechanisms that revolutionized natural language processing and enabled modern LLMs.
- **Advances in LLM Text Generation:** Models such as GPT-4, Gemini, and LLaMA demonstrate state-of-the-art performance in contextual text generation, summarization, and tone adaptation tasks.
- **Prior Email Automation Approaches:** Earlier systems relied on rule-based templates and keyword matching, providing limited contextual understanding and personalization capabilities.
- **Limitations of Template-Based Systems:** Template-driven approaches lack adaptability to diverse communication styles, fail to capture nuanced context, and produce generic, impersonal responses.
- **Prompt Engineering Techniques:** Recent research highlights the importance of structured prompt design in optimizing LLM output quality for domain-specific applications such as professional email composition.

4. Proposed Model



5.Discussion About Methodology & Algorithm

Email Processing Workflow



6.Conclusion & Future Work

Conclusion:

- LLM-based email automation systems can significantly reduce the manual effort and time invested in professional email drafting and communication.
- The proposed architecture demonstrates effective integration of prompt construction, LLM processing, and post-processing layers to generate contextually appropriate and personalized email content.

Future Work:

- Multilingual email generation to support global enterprise communication.
- Improved personalization through advanced user preference learning and behavioral analysis.
- Enterprise system integration with CRM, ERP, and collaboration platforms.
- Context-aware AI assistants capable of proactive email management and intelligent scheduling.

7. Research Paper Progress

- **Architecture Design:** The system architecture comprising the user interface layer, backend processing layer, LLM processing layer, post-processing layer, and data intelligence layer has been finalized.
- **Literature Review Completion:** A comprehensive review of transformer-based models, prompt engineering techniques, and prior email automation systems has been completed and documented.
- **Prototype System Development:** An initial prototype demonstrating core email generation, summarization, and tone rewriting capabilities has been developed and tested.
- **Experimental Evaluation:** Benchmarking and evaluation metrics for measuring productivity improvements, response quality, and user satisfaction are being defined for systematic assessment.

8.References

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